COLLEGE NAME : DONBOSCOCLEGE (CO-ED), YELAGIRI HLLS

COLLEGE CODE : 358

DEPARTENT : BCA

TITLE : Courier Management System (CMS)

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ABSTRACT

Courier Management System (CMS)

Courier and logistics organizations require reliable systems to handle shipments, monitor deliveries, and ensure that packages are delivered to customers on time. However, many courier services still depend on manual procedures or multiple disconnected systems for managing shipment booking, delivery tracking, and billing operations. These traditional approaches often lead to delays, data inaccuracies, and limited visibility for customers, which negatively affects operational efficiency. In addition, it becomes challenging for companies to properly monitor deliveries and maintain accurate records.

The **Courier Management System (CMS)** is a cloud-based solution developed using Salesforce to address these challenges. The system allows courier companies to manage all operational activities from a single platform, including shipment registration, package tracking, delivery agent allocation, status updates, and billing management. By centralizing these processes, the system improves data organization and ensures that information is easily accessible. It also enhances employee productivity by minimizing manual tasks.

The application is designed using custom objects such as **Shipment, Customer, Delivery Agent, and Delivery Status** to store and manage courier-related information efficiently. Automated workflows are implemented to update shipment status, assign delivery personnel, and handle billing processes automatically. This automation reduces manual effort, increases accuracy, and improves coordination among different departments within the courier service.

Furthermore, the system provides analytical reports and dashboards that help managers evaluate delivery performance and monitor shipment activities in real time. These reports enable companies to analyze operational efficiency and identify areas that require improvement. Managers can quickly access essential information and make informed decisions. By transforming manual processes into an automated cloud-based platform, the Courier Management System significantly improves operational efficiency while also enhancing customer satisfaction through accurate and timely delivery updates.

PROBLEM STATEMENT

Courier and logistics companies manage a large volume of shipments every day and must ensure that parcels reach customers safely and within the expected time. Effective shipment monitoring, delivery coordination, and clear communication with customers are essential for maintaining high service standards. However, many courier organizations still rely on manual procedures or multiple unconnected systems to handle shipment registration, parcel tracking, and delivery management. These outdated approaches create several operational difficulties and decrease overall efficiency.

A major challenge faced by courier companies is the absence of a centralized platform to control and manage courier activities. Branch managers often struggle to organize shipment information, monitor delivery progress, and allocate delivery agents efficiently. Since the data is frequently stored in different locations or recorded manually, maintaining accurate, consistent, and up-to-date records becomes difficult.

• **I am:** Branch Manager
• **I'm trying to:** Manage shipments and delivery activities effectively
• **But:** Shipment tracking is handled manually
• **Because:** There is no unified system to store and manage shipment details
• **Which makes me feel:** Overwhelmed when dealing with a high volume of deliveries

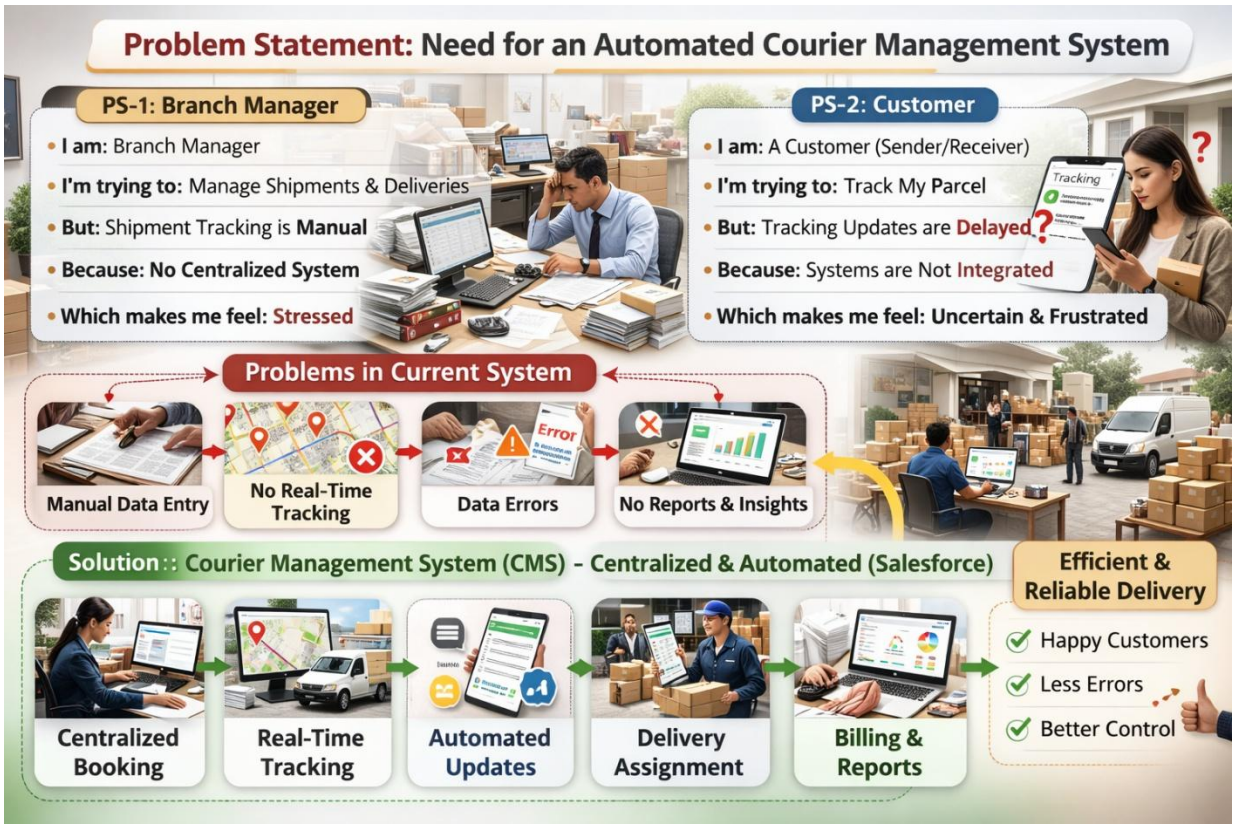
Customers also encounter difficulties when attempting to track their parcels. Because shipment updates are entered manually or maintained across separate systems, tracking information is often delayed or incomplete. This lack of transparency lowers customer trust in the courier service and makes it harder for them to know the exact status of their packages.

• **I am:** A customer sending or receiving a parcel
• **I'm trying to:** Check the status and location of my parcel
• **But:** Tracking updates are slow or missing
• **Because:** Courier systems are not integrated and lack automation
• **Which makes me feel:** Confused and frustrated about the delivery progress

Another significant problem is the lack of automation and analytical capabilities in courier management. Manual data entry increases the likelihood of mistakes and makes it challenging to generate reports or evaluate delivery performance. As a result, managers find it difficult to monitor operational efficiency or identify issues in the delivery workflow.

To address these challenges, an efficient and automated system is required to manage courier services effectively. The **Courier Management System (CMS)** provides a centralized cloud-based platform developed using Salesforce. It enables organizations to handle shipment booking, parcel tracking, delivery agent assignment, automated status updates, billing, and reporting within a single system. By integrating these operations, the system improves operational efficiency, increases customer satisfaction, and supports better decision-making through accurate and real-time data.

PROBLEM STATEMENT FOR COURIER MANAGEMENT SYSTEM



PROPOSED SOLUTION

To overcome the identified challenges, the proposed solution is the development of a **Courier Management System (CMS)**, a cloud-based courier and logistics management application developed using **Salesforce**. The system is designed to provide a **structured, scalable, and automated platform** for managing courier operations such as shipment booking, parcel tracking, delivery agent assignment, billing, and reporting. By centralizing courier data in one system, the application improves operational efficiency, ensures accurate data management, and provides real-time visibility of shipment status across multiple branches.

Customer Object

This object stores personal and contact details of customers who send or receive parcels. It helps maintain proper customer records and track shipment history.

The Customer object stores details such as:

• Customer Name
• Email Address
• Phone Number
• Address
• Customer Type (Regular / Business)
• City
• State
• Pincode
• Active Status

Shipment Object

This object manages information related to courier shipments and parcel details. It helps track the movement and delivery status of each shipment.

The Shipment object includes details such as:

• Tracking Number
• Customer
• Source Address

• Destination Address
• Package Weight
• Shipment Price
• Shipment Status (Booked, In Transit, Out for Delivery, Delivered, Cancelled)
• Expected Delivery Date
• Assigned Delivery Agent
• Shipment Type (Domestic / International)

Delivery Update Object

This object records the real-time status updates for each shipment during the delivery process.

The Delivery Update object captures details such as:

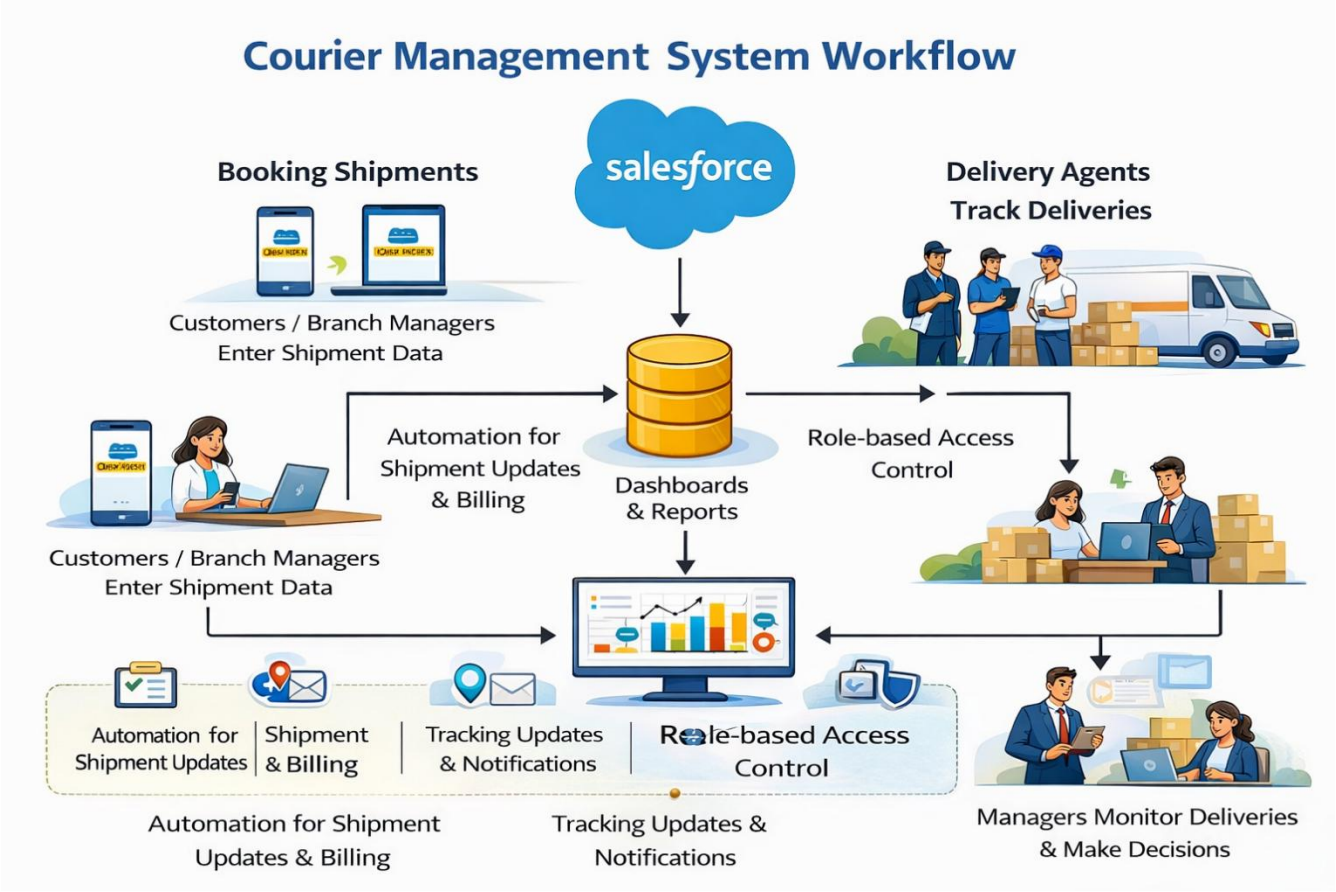
• Update Number
• Shipment
• Status (In Transit, Out for Delivery, Delivered, Delayed)
• Update Date and Time
• Remarks

A **Master-Detail relationship** is established between **Shipment (Master)** and **Delivery Update (Detail)**. This ensures strong data integrity and allows the system to maintain proper tracking records for every shipment. If a shipment record is deleted, all related delivery updates are automatically removed, maintaining consistency in the system.

The application leverages **Salesforce's reporting and analytics engine** to generate real-time insights such as shipment status summaries, agent performance reports, and branch-wise revenue analysis.

Thus, the **Courier Management System (CMS)** transforms manual courier management processes into a **structured, automated, and intelligent logistics management system**, improving operational efficiency, delivery tracking accuracy, and customer satisfaction.

WORK FLOW DIAGRAM



SUMMARY

The **Courier Management System (CMS)** is a cloud-based application developed using Salesforce to improve courier and logistics operations. It helps organizations manage shipment booking, parcel tracking, delivery agent assignment, and billing in a centralized system. By replacing manual processes with automation, the system improves operational efficiency and reduces errors. It also ensures better coordination between different branches and departments.

The system uses several custom Salesforce objects such as Customer, Shipment, Delivery Agent, Branch, Delivery Update, and Invoice to manage courier data effectively. These objects store important information related to customers, parcels, delivery agents, and billing. Relationships between these objects ensure proper data organization and consistency. This structured data model helps manage courier operations smoothly.

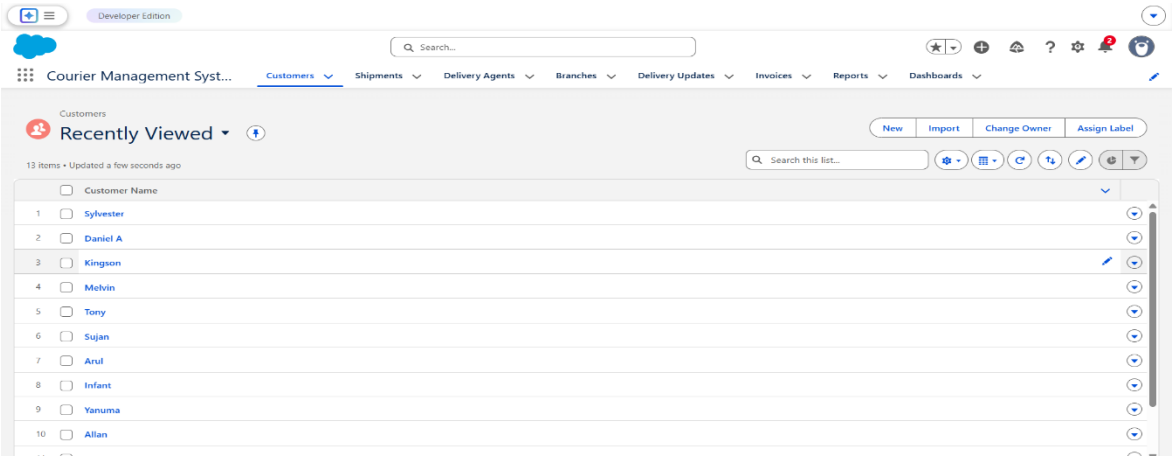
Automation features such as flows, validation rules, and automated status updates help reduce manual work and ensure accurate data entry. Real-time shipment tracking allows customers and managers to monitor parcel movement and delivery progress. The system also provides automated billing and notifications, improving communication and service reliability.

Additionally, reports and dashboards provide valuable insights into shipment status, delivery performance, and branch revenue. Managers can analyze operational data and make better business decisions. These visual tools help track courier activities and identify areas that require improvement.

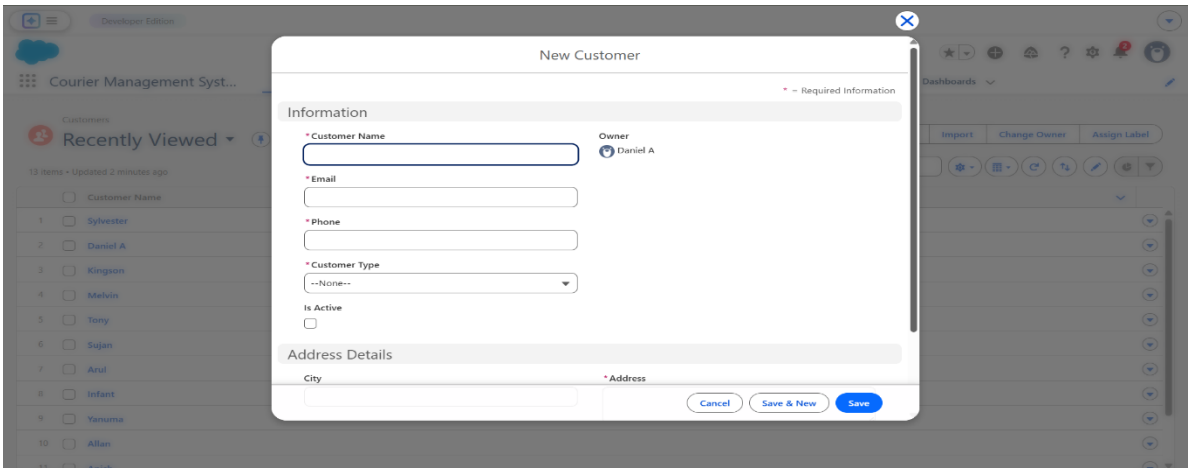
Overall, the Courier Management System transforms traditional courier management into a modern digital platform. It improves transparency, enhances operational efficiency, and increases customer satisfaction through real-time tracking and automation. By using Salesforce technology, the system supports scalable courier operations and helps organizations manage deliveries more effectively.

SCREENSHOTS

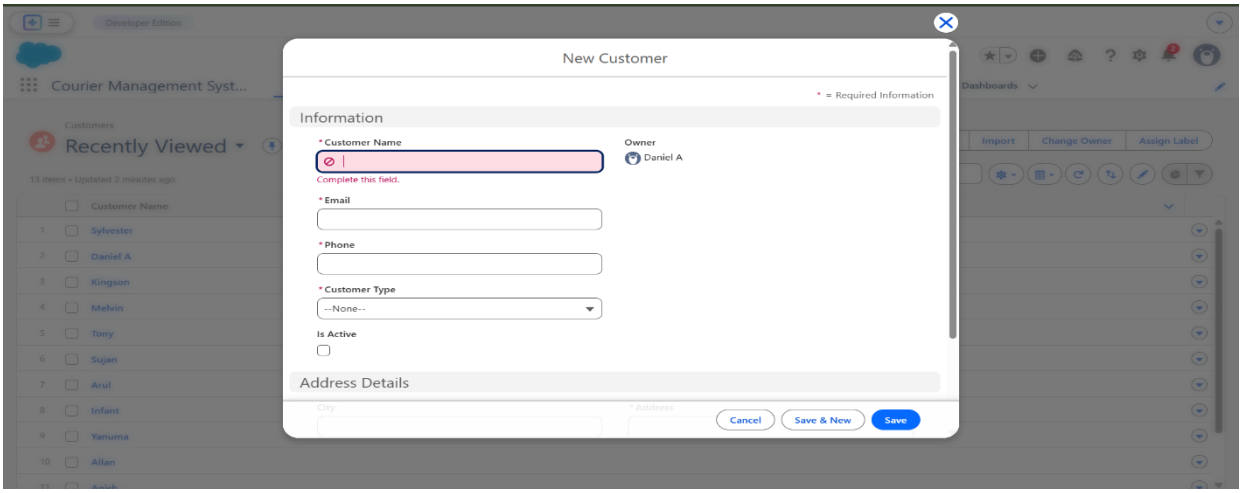
CUSTOMER LIST VIEW PAGE IN COURIER MANAGEMENT SYSTEM



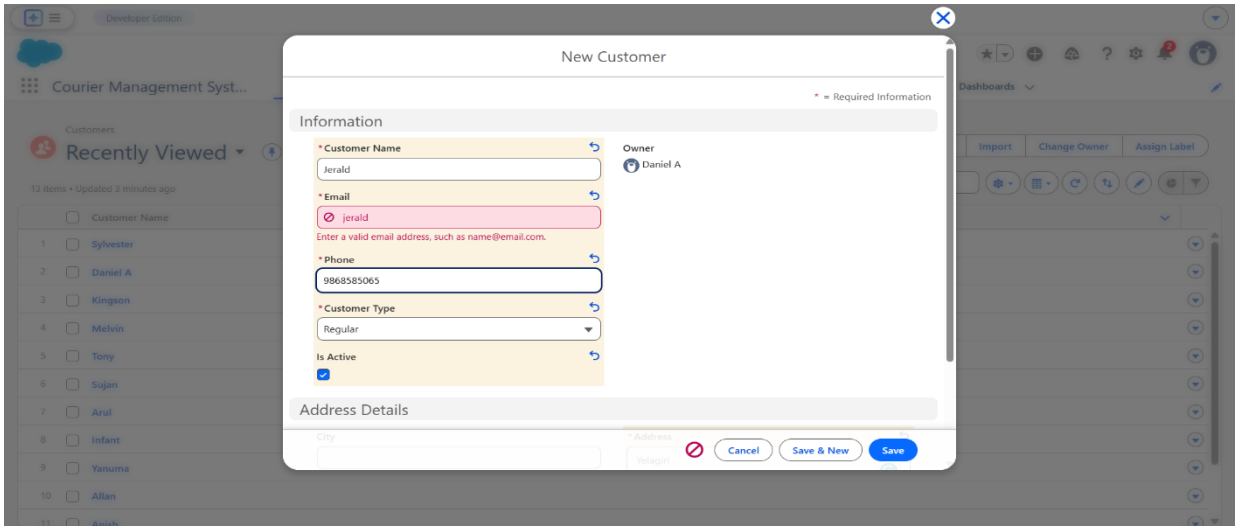
NEW CUSTOMER CREATION FORM IN CUSTOMER ACCOUNTS PAGE



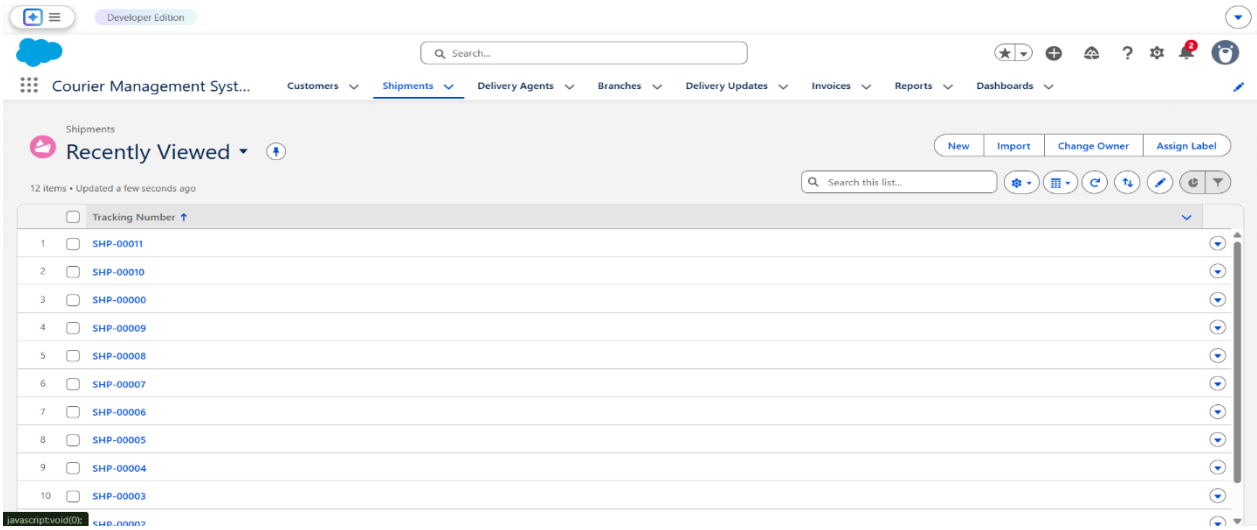
CUSTOMER NAME REQUIRED FIELD VALIDATION ERROR IN CUSTOMER ACCOUNTS :



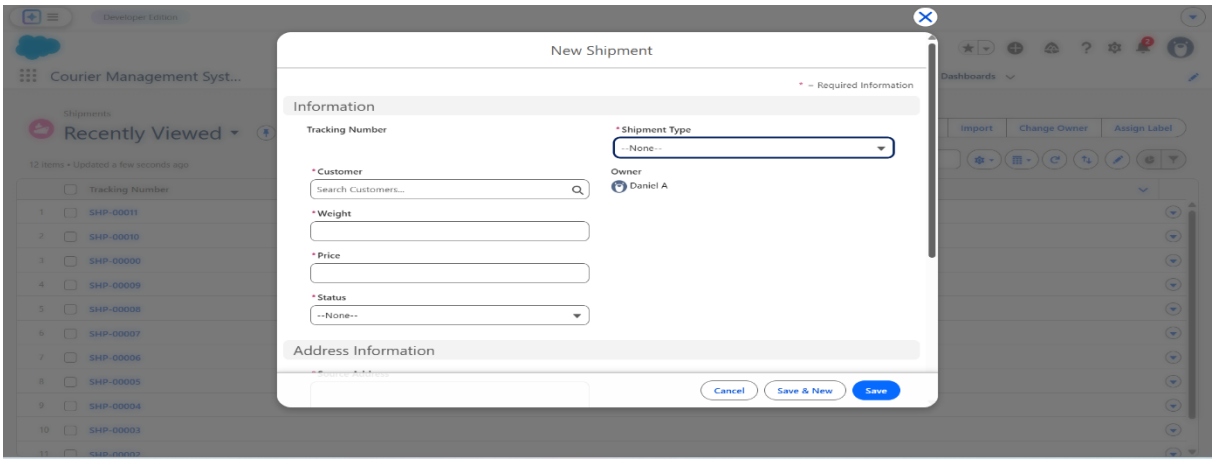
INVALID EMAIL FORMAT ERROR IN CUSTOMER ACCOUNTS PAGE



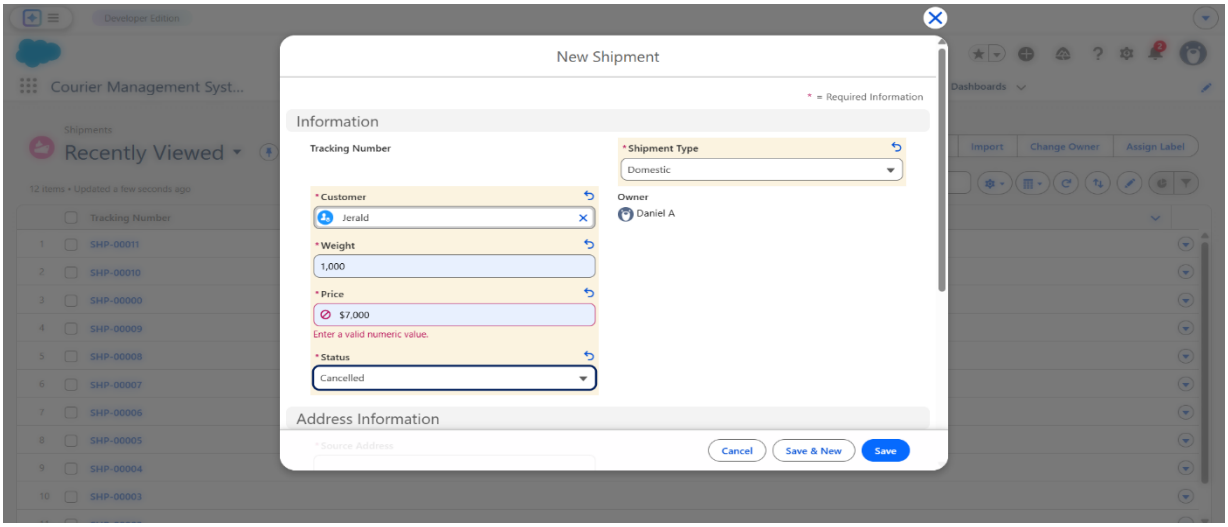
SHIPMENT LIST VIEW PAGE IN SHIPMENTS MODULE



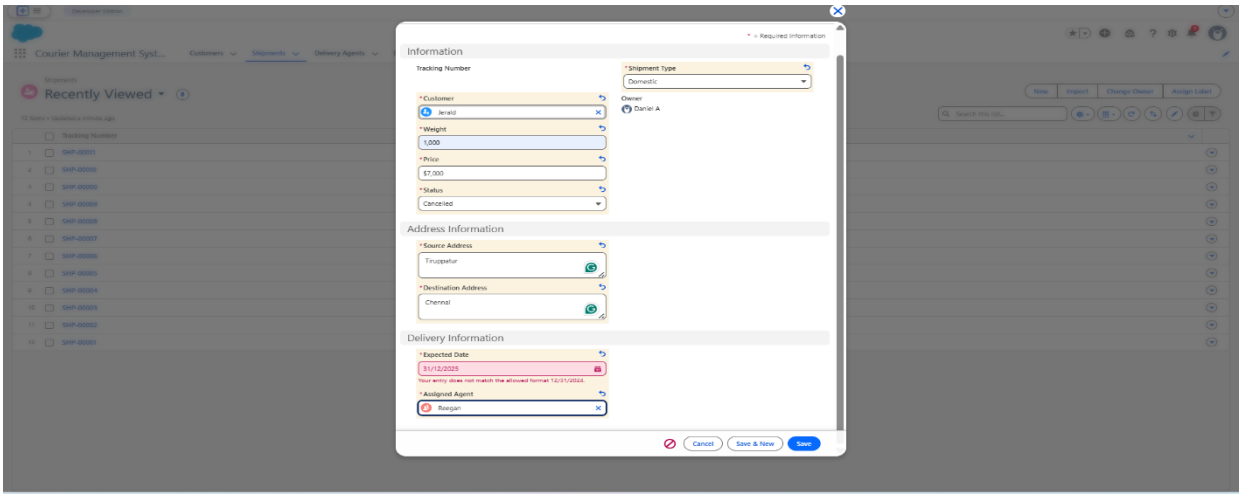
NEW SHIPMENT CREATION FORM IN SHIPMENTS PAGE



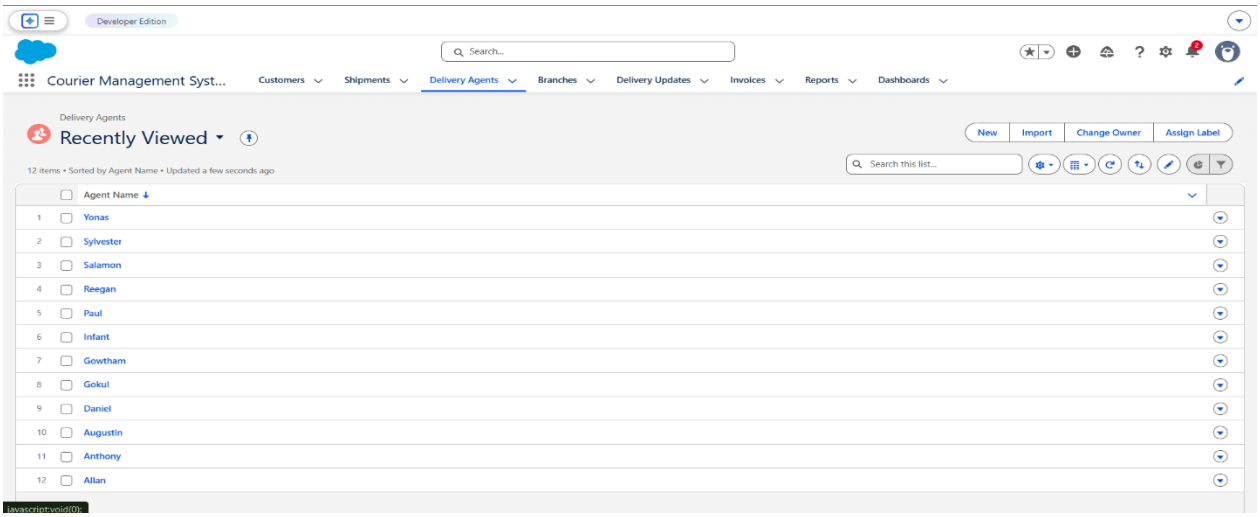
PRICE FIELD INVALID VALUE ERROR IN SHIPMENTS PAGE



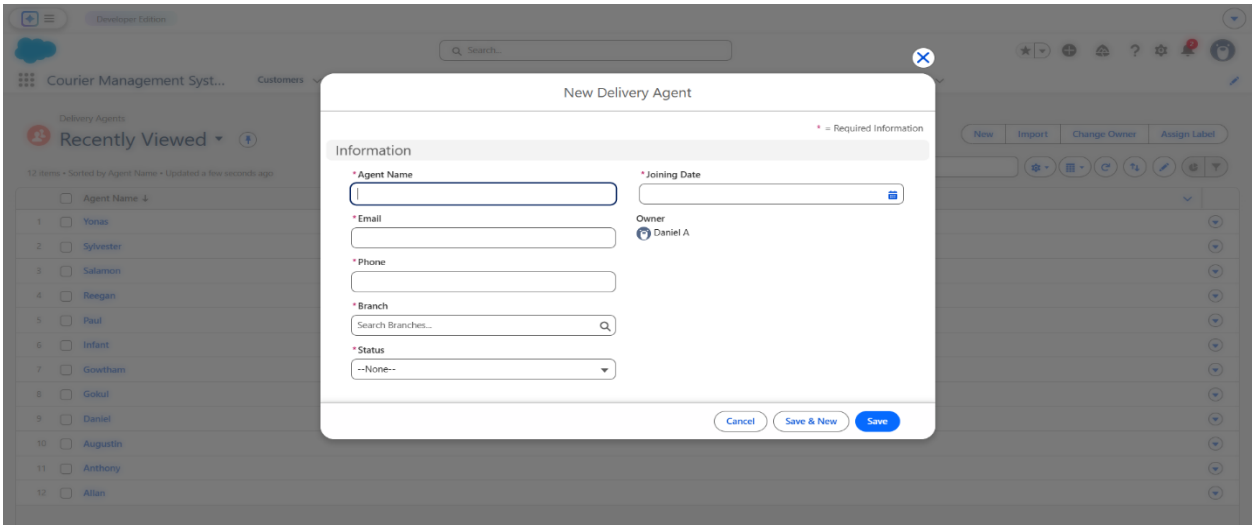
DELIVERY INFORMATION REQUIRED FIELD ERROR IN SHIPMENTS PAGE



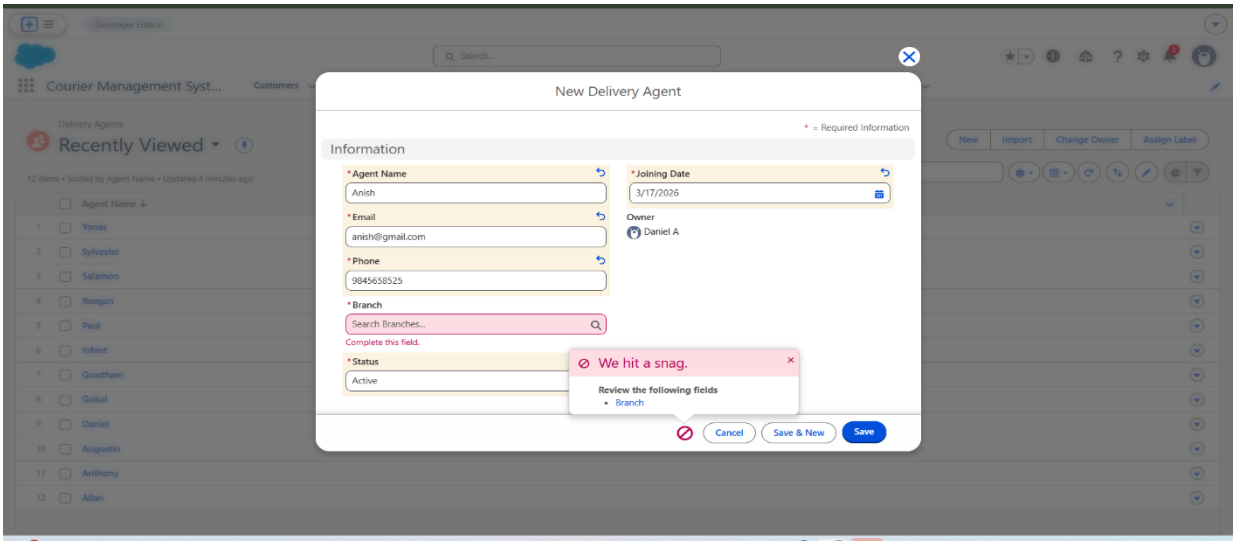
DELIVERY AGENT LIST VIEW PAGE IN DELIVERY AGENTS MODULE



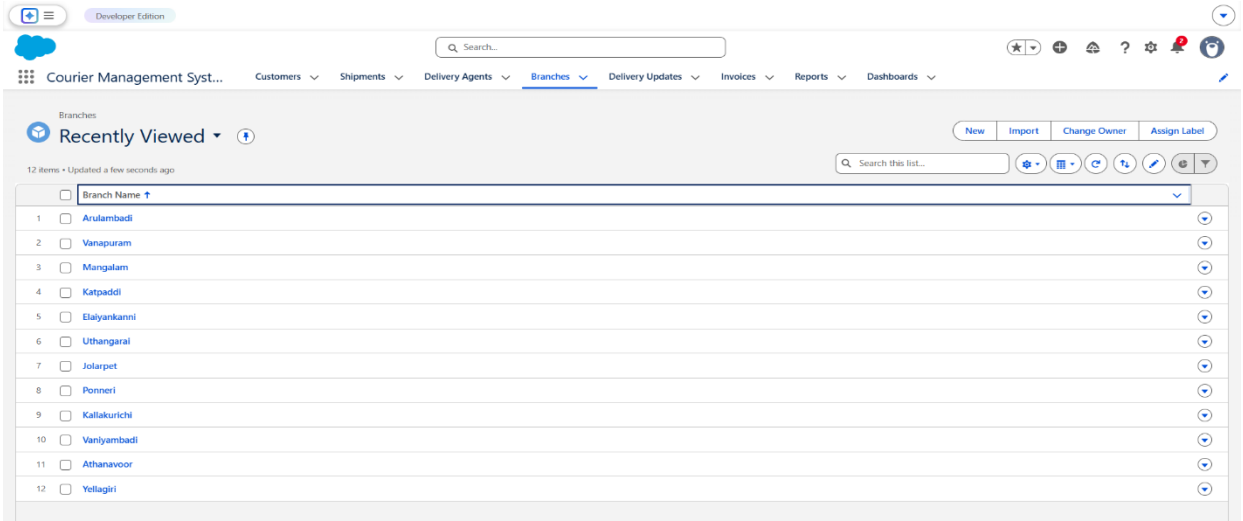
NEW DELIVERY AGENT CREATION FORM IN DELIVERY AGENTS PAGE



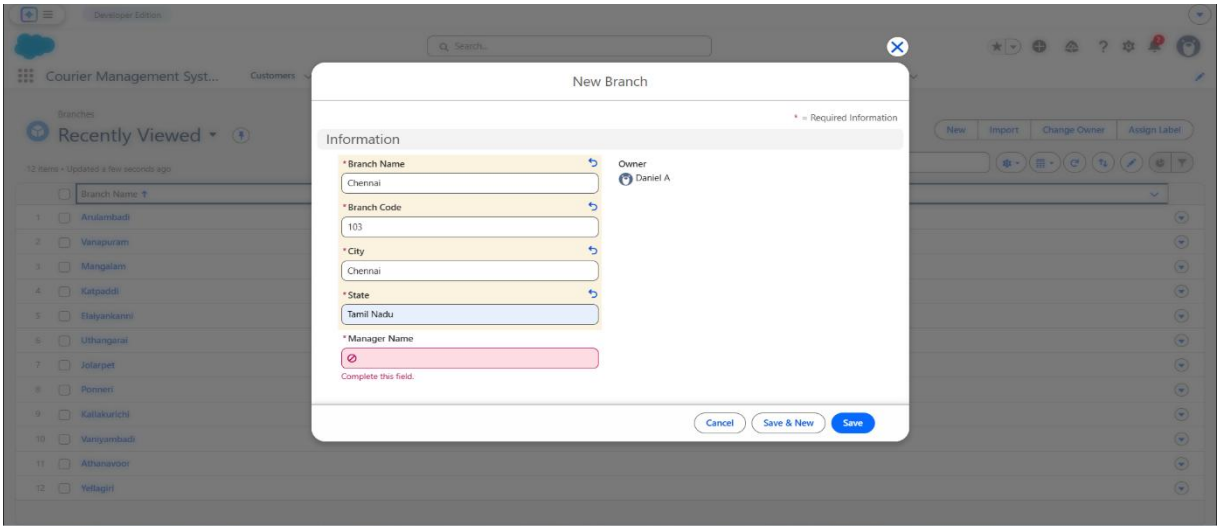
BRANCH FIELD NOT SELECTED ERROR IN DELIVERY AGENTS PAGE



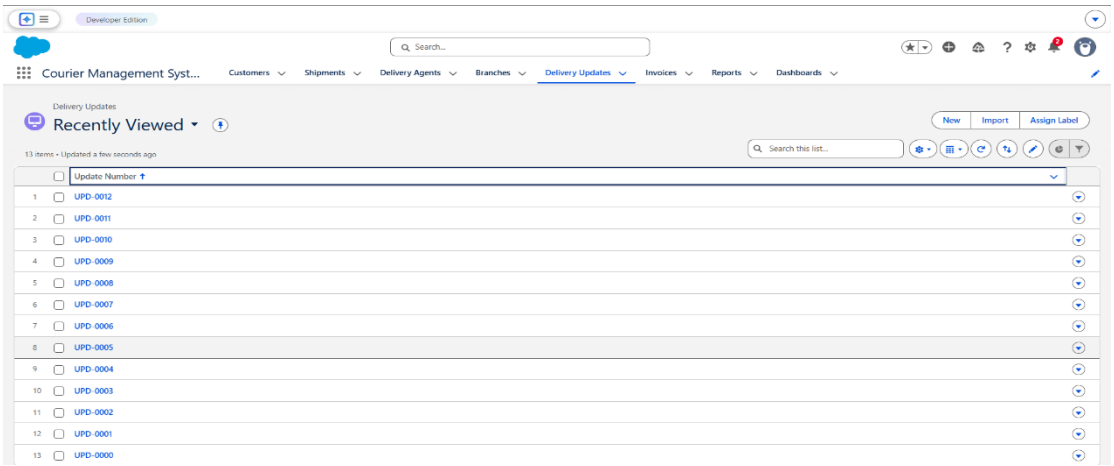
BRANCH LIST VIEW PAGE IN BRANCHES MODULE



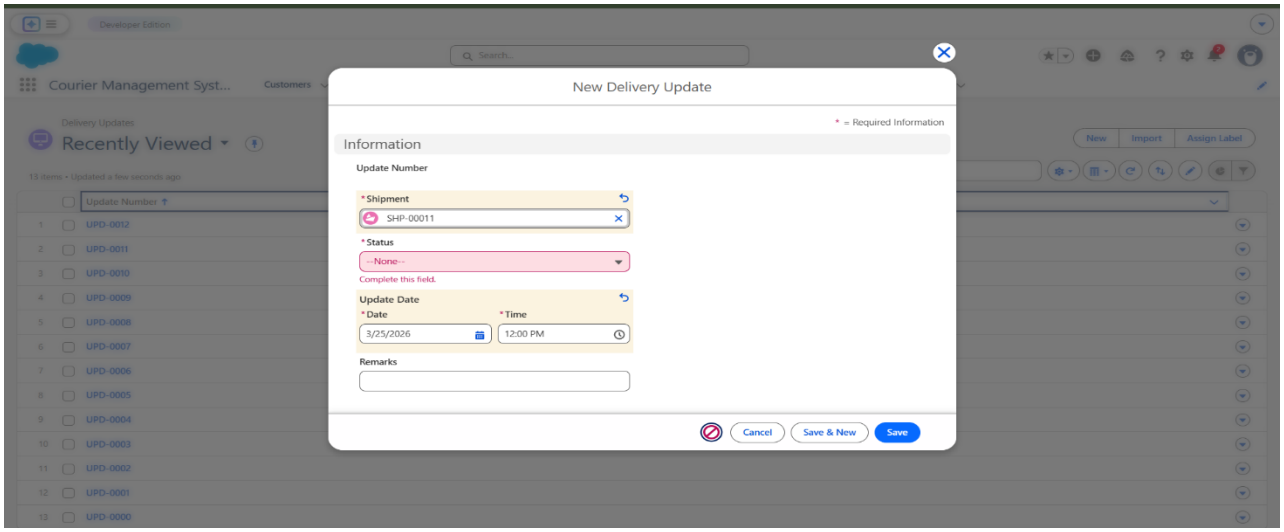
MANAGER NAME REQUIRED FIELD ERROR IN BRANCH CREATION PAGE



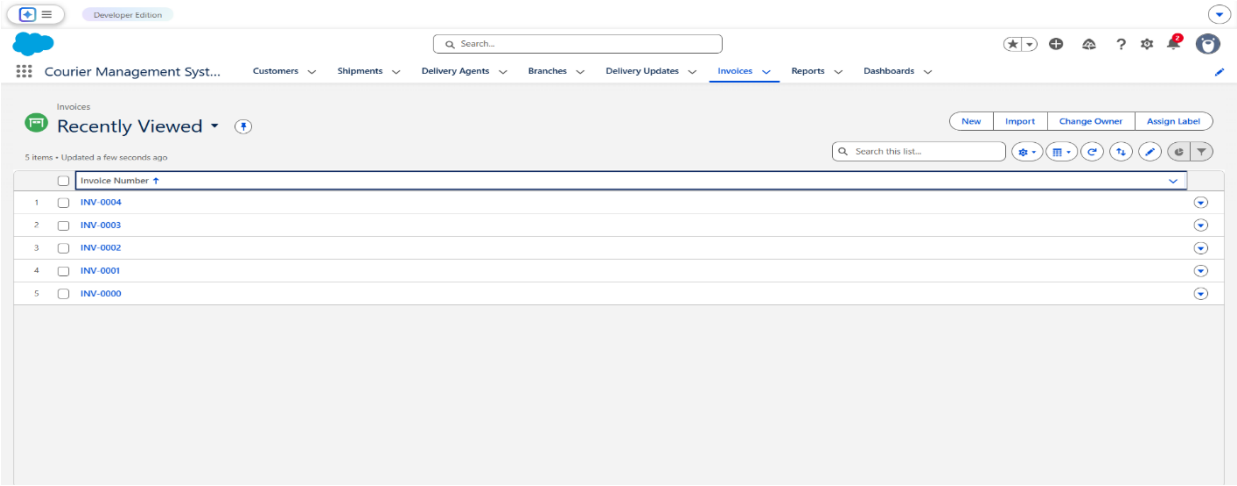
DELIVERY UPDATE LIST VIEW PAGE IN DELIVERY UPDATES MODULE



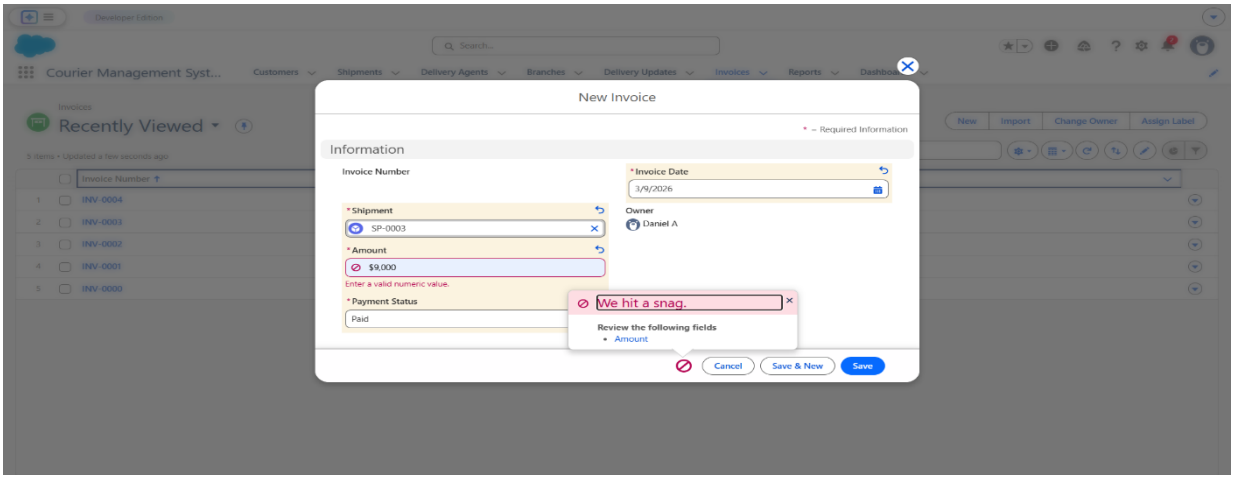
STATUS REQUIRED FIELD ERROR IN DELIVERY UPDATE PAGE



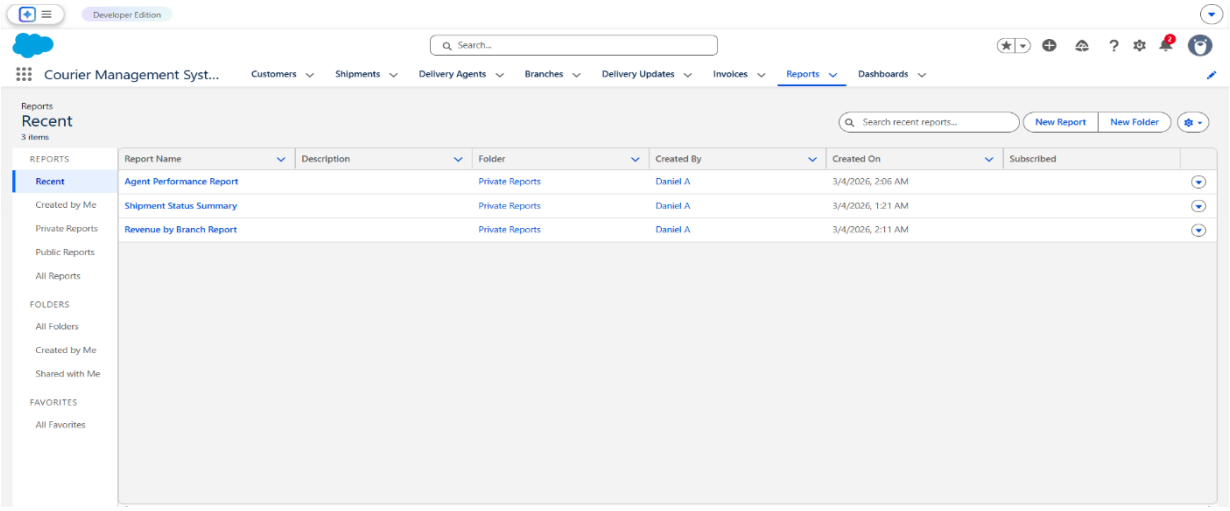
INVOICE LIST VIEW PAGE IN INVOICE MODULE



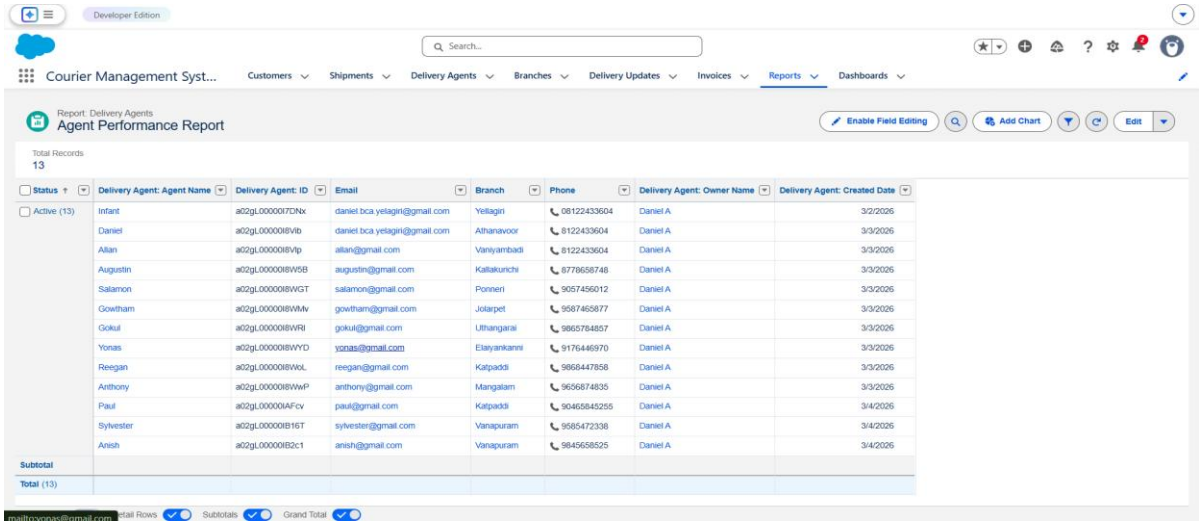
AMOUNT FIELD INVALID VALUE ERROR IN INVOICE CREATION PAGE



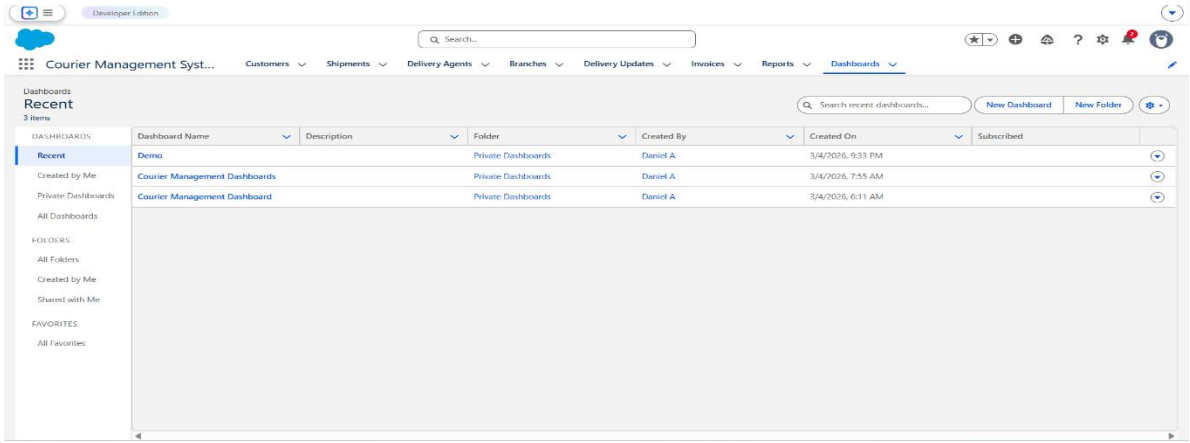
REPORTS LIST VIEW PAGE IN REPORTS MODULE



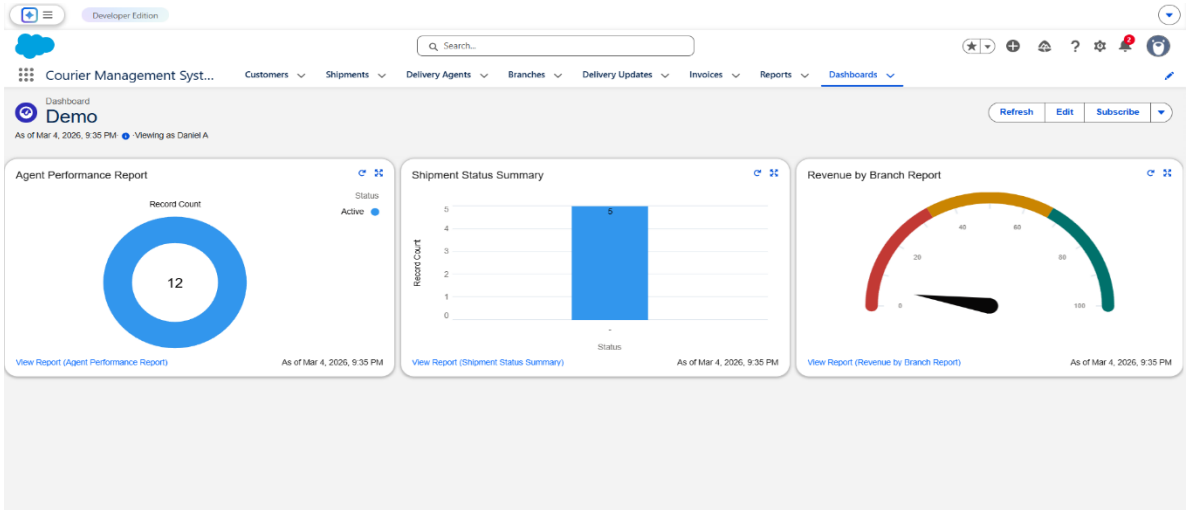
DELIVERY AGENT PERFORMANCE REPORT PAGE



DASHBOARD LIST VIEW PAGE IN DASHBOARD MODULE



COURIER MANAGEMENT SYSTEM ANALYTICS DASHBOARD



CONCLUSION

The **Courier Management System (CMS)** is designed to improve the efficiency of courier and logistics operations by using the Salesforce platform. It replaces traditional manual methods with an automated and centralized system. This helps organizations manage shipments, customers, and delivery agents more effectively.

The system provides features such as **shipment booking, delivery tracking, agent assignment, and billing management**. These functions help courier companies monitor parcel movement and delivery progress in real time. As a result, operational delays and manual errors are reduced.

By using **custom objects, relationships, and automation tools**, the system ensures that courier data is organized and updated accurately. Validation rules and automated workflows help maintain data quality and reduce manual effort. This improves the reliability of courier operations.

The **reports and dashboards** generated by the system provide useful insights into shipment status, delivery performance, and revenue. Managers can analyze this information to monitor business performance and make better decisions. These analytics help improve overall operational efficiency.

Overall, the Courier Management System successfully transforms manual courier processes into a **structured and automated digital system**. It enhances operational efficiency, improves customer satisfaction, and supports better management of courier services. The system demonstrates how Salesforce technology can be used to optimize logistics operations.